David Jovany Marin Holguin

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EDUCATION

May 2027

University of Kentucky **Lexington, KY**

Bachelor of Science in Biomedical Engineering Major GPA: 3.6

INTERNSHIP EXPERIENCE

Johnson & Johnson Ethicon **Cincinnati, Ohio**

Design Engineering Co-op January 2026- May 2026

- Supported lifecycle management for 10 harmonic and advanced bipolar surgical devices by applying design controls and developing SolidWorks models, ensuring regulatory compliance and reducing design change requests across 15+ projects.
- Led verification testing, updated test methods, and performed statistical analysis across eight product families, supporting data-driven decisions that improved product reliability and manufacturability.
- Collaborated cross-functionally with manufacturing, quality, design, and cost-savings teams to minimize production line downtime and improve operational efficiency.
- Conducted IP and prior-art research to evaluate existing technologies and product differentiation, presenting findings and design implications to cross-functional stakeholders.

Atlantis: Clinical Immersion **Rome, Italy**

Clinical Shadowing- Orthopedic Rotation May 2025- June 2025

- Completed 100+ clinical shadowing hours observing medical procedures and device usage, strengthening understanding of clinical applications and patient outcomes.

Shriners Medical Hospital **Lexington, Kentucky**

PT/OT Intern & Prosthetics Spring 2023 & Fall 2025

- Implemented evidence-based therapy protocols and monitored patient progress through motion analysis, contributing to enhanced treatment outcomes for 12+ patients.
- Observed prosthetic adjustments and biomaterial selection to evaluate device fit, mobility, and patient outcomes while gaining exposure to V&V and risk-management principles.

UNIVERSITY PROJECTS

Lexington, Kentucky

Senior Design- Adaptive Pressure Distribution Orthotic Insert Fall 2026

- Developing a low-profile orthotic platform using unidirectional anisotropic carbon fiber, pressure-relieving materials, and replaceable sensing patches to address abnormal plantar loading.
- Translate sponsor and clinician needs into engineering requirements, acceptance criteria, material selections, and verification strategies while considering biomechanics, usability, and manufacturability.

Design & Development of Knee Orthotic Fall 2025

- Collaborated to design and develop an innovative knee orthotic for athletes, integrating electrical muscle therapy for passive recovery.
- Integrated design thinking to conduct prototype testing and evaluation, analyzing performance and clinical usability for athlete recovery.
- Performed CAD modeling in SolidWorks, conducted multiple design iterations, and selected materials based on user feedback to optimize functionality and manufacturability.

ECG Data Analysis & Respiratory Rate Detection Fall 2025

- Manufactured MATLAB algorithms to receive raw ECG data and calculate user respiratory rate derived from R-peak detection using engineering analysis, validation, and documented results in a formal technical report.

CERTIFICATIONS AND TECHNICAL SKILLS

- **Design & Simulation:** SolidWorks, Siemens NX, Onshape, MATLAB, Minitab
- **Medical Device Development:** V&V, test methods, design controls, FMEA, risk management, statistical analysis
- **Documentation:** Test protocols, validation reports, laboratory notebooks, documentation control
- **Certifications:** Toyota True LEAN, NOCTI Lab Safety, Harvard Medical Pro Pharmacology: Drug Delivery, AAPPL Seal of Biliteracy: Spanish & English